PH.48111 **FALL 2026**

Autonomous Theory Construction Laboratory

Lecture 11: Cross-Domain Theory Grammars

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Overview

The same loop interface is specialized for Hamiltonian, transport, and statistical models through domain-specific constraints.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$Loop = (Registry, Grammar, Evaluator, Acceptor, Log)$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$G_{QM} \vdash H = H^{\dagger}, p \geq 0$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$G_{thermal} \vdash \partial_t \rho + \nabla \cdot J = S - D$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Yun, ch. 11
- Cross-Domain Grammar Atlas
- QM registry profile